

AGENT-BASED MODELING · TECHNOLOGY IN SOCIETY

Dark Patterns as a Value-Destroying Strategy

An Agent-Based Network Model of Trust, Social Contagion, and Platform Sustainability

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Agenda

1. **The problem** — dark patterns are everywhere, and they work
2. **The gap** — nobody has modeled the *long-run* systemic effect
3. **The model** — a network-based agent-based simulation
4. **The results** — six years of simulated trust erosion
5. **The takeaways** — tipping points, vulnerability, and policy

Central question: if dark patterns boost short-term conversions, do they actually pay off for a platform over years — or do they quietly destroy it?

PART 1

The Problem

Dark patterns are the norm, not the exception

1,818

DARK PATTERN INSTANCES FOUND
ACROSS 11,000 SHOPPING SITES (MATHUR
ET AL. 2019)

~95%

OF 240 POPULAR MOBILE APPS CONTAIN
AT LEAST ONE DARK PATTERN (DI
GERONIMO ET AL. 2020)

~2–4×

INCREASE IN UNWANTED SIGN-UPS FROM
MILD → AGGRESSIVE DARK PATTERNS
(LUGURI & STRAHILEVITZ 2021)

Deceptive interfaces exploit framing effects, scarcity illusions, social proof, and anchoring — steering users into purchases, data disclosure, and subscriptions they would not otherwise choose.

The research gap

Existing research is **predominantly static**: cross-sectional detection and classification, or one-shot experiments measuring immediate response.

What's missing:

- How dark patterns interact with **social networks** over time
- How harm **accumulates** and **spreads** via word-of-mouth
- Whether "profitable" dark patterns are actually **sustainable**
- *At what intensity* systemic harm begins

No prior model couples a platform's strategic design choices, heterogeneous user behavior, **and** the resulting trust–economy feedback loop in one computational framework.

This paper's contribution

1. A network-based ABM

Simulates 6 years of trust, harm, churn, word-of-mouth, reputation, and platform economics under dark-pattern deployment.

2. Four formal tipping points

Quantitative detectors for qualitative regime shifts in platform health.

3. Systematic scenario analysis

100 random seeds × 7 scenarios reveal a **non-linear, threshold-dependent** relationship between intensity and sustainability.

Headline finding: **no safe deployment level exists** — even mild dark patterns cross regime boundaries.

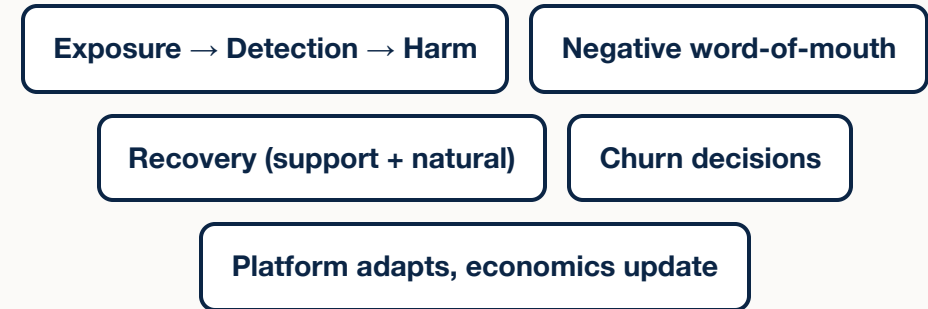
PART 2

The Model

Model overview

- **500** heterogeneous user agents
- Connected via a **small-world social network** (Watts–Strogatz)
- **1** platform agent deploying up to 3 dark patterns
- **1 step = 1 week** → 312 steps ≈ **6 years**
- Built in **Mesa** (Python ABM framework)

Each weekly step:



Three user archetypes

Type	Share	Key traits
Naive	50%	High baseline trust, low digital literacy, low manipulation sensitivity, high switching cost , elevated trust resilience
Skeptic	30%	Moderate trust, high digital literacy, higher manipulation sensitivity, lower switching cost
Activist	20%	Moderate trust, highest manipulation sensitivity, lowest switching cost, high social activity

Traits are sampled from Beta(5,5) distributions scaled to type-specific ranges — bell-shaped variation around each type's midpoint, not a single universal distribution.

Three dark patterns modeled

Pattern	Models	Detectability	Base harm
Forced Trial	Forced conversions / upsells after a "free" trial	Moderate	Moderate
Hard Cancellation	Obstruction-based subscriber retention	Highest	Moderate
Drip Pricing	Hidden fees revealed late in checkout	Moderate	Highest

All three share a single global **dark-pattern intensity** dial (0–1). Trust loss and harm accumulate per exposure, with per-step caps and a 3-exposure "habituation" ramp so shocks build gradually rather than instantly.

Four formal tipping points

Each requires **5 consecutive steps** past threshold to trigger — filtering noise from genuine regime shifts.

Tipping point	Trigger condition
Trust Collapse	Mean trust of active users ≤ 0.50
Social Contagion	Negative word-of-mouth rate ≥ 0.22
Churn Cascade	Cumulative churn ≥ 0.35
Extractive Divergence	Revenue gap $\geq 20\%$ of short-term revenue and churn ≥ 0.15

These operationalize qualitative "platform health" shifts that are hard to observe empirically but emerge clearly in simulation.

Experimental setup

500

SIMULATED USERS

312

WEEKLY STEPS (~6 YEARS)

100

RANDOM SEEDS PER
SCENARIO

7

SCENARIOS COMPARED

Scenarios: Control (no dark patterns) · Low (0.20) · Medium (0.40) · High (0.80) — all three patterns active — plus three **single-pattern** scenarios at intensity 0.50 to isolate each pattern's individual effect.

All results regenerated end-to-end from an open replication package; unit-tested exposure-harm pipeline and tipping-point detectors.

PART 3

Results

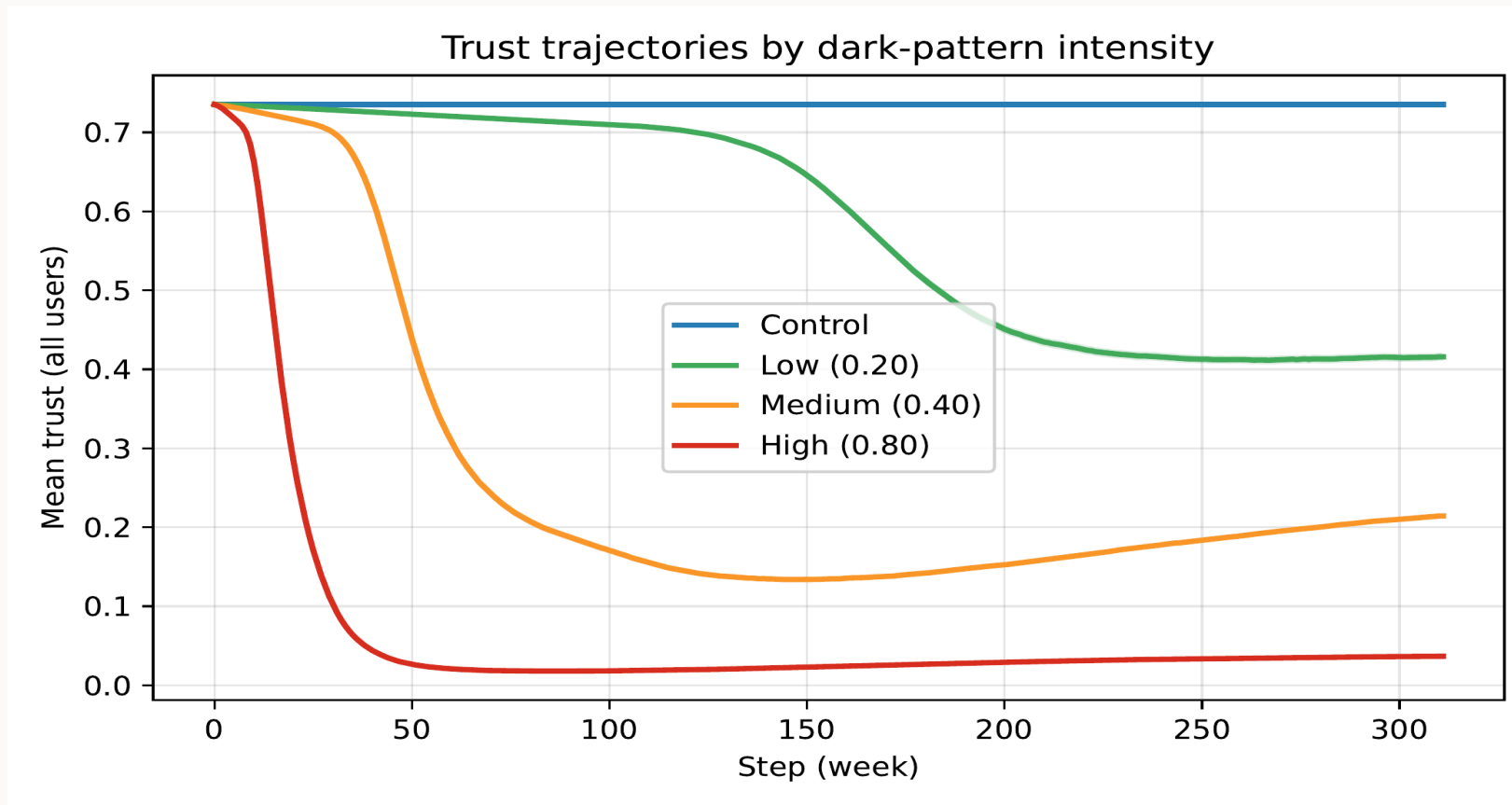
Outcomes across intensity levels

TABLE 1 · 312 STEPS · N = 500 · 100 SEEDS · MEAN ± 95% CI

Metric	Control	Low (0.20)	Medium (0.40)	High (0.80)
Cumulative churn	8.2 ± 0.2%	17.6 ± 0.3%	53.5 ± 0.5%	88.0 ± 0.3%
Mean trust (all)	0.735	0.416	0.214	0.037
Platform reputation	74.3	46.8	18.4	2.0 (floor)
Cumulative revenue	654,255	624,184	452,232	281,916
Opportunity cost	29,079	59,151	231,103	401,419
Tipping points (of 4)	0.00	1.95	4.00	4.00

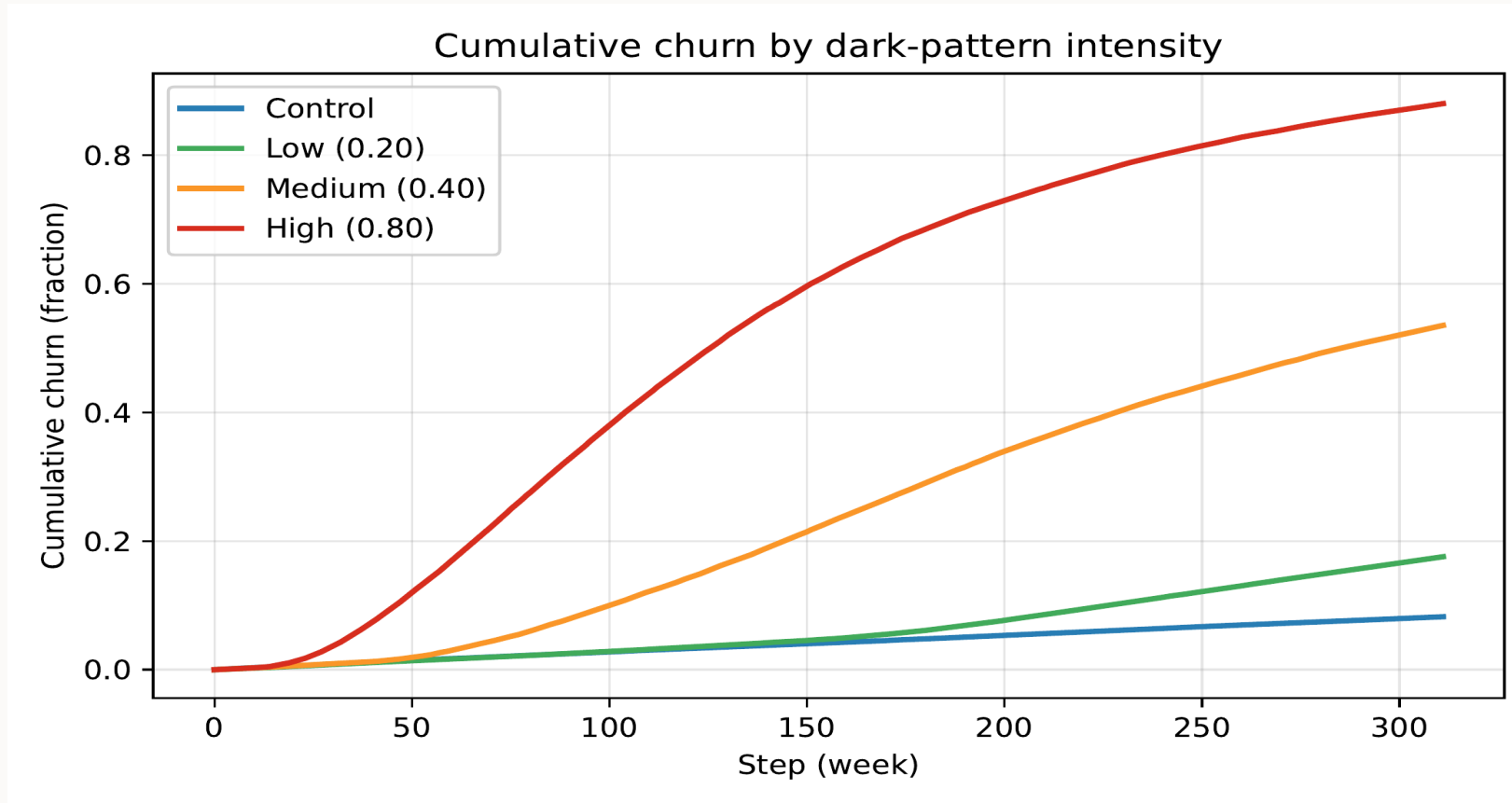
Even **low** intensity (0.20) crosses Trust Collapse and Extractive Divergence — while revenue looks almost untouched.

Trust collapses well before revenue visibly suffers



Mean trust (all users) over time by dark-pattern intensity; shaded bands = 95% CI across 100 seeds.

Churn accelerates sharply with intensity



Cumulative churn over time by dark-pattern intensity (mean $\pm$ 95% CI).

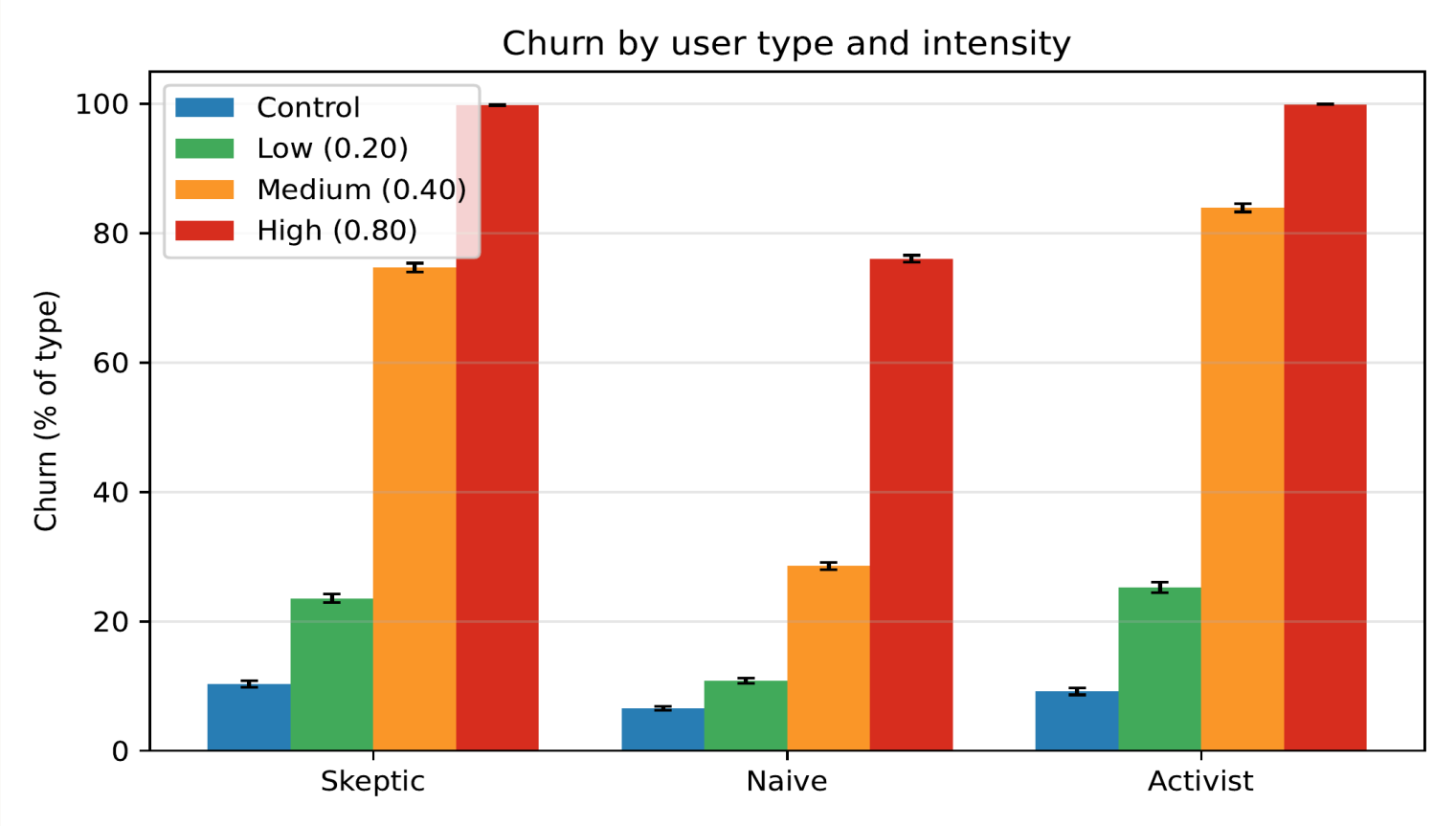
Who leaves — and who stays trapped

TABLE 2 · CHURN BY USER TYPE · 100 SEEDS

User type	Control	Low (0.20)	Medium (0.40)	High (0.80)
Skeptic (N=150)	10 ± 1%	24 ± 1%	75 ± 1%	100%
Naive (N=250)	7 ± 0%	11 ± 0%	29 ± 1%	76 ± 1%
Activist (N=100)	9 ± 1%	25 ± 1%	84 ± 1%	100%

Skeptics and activists exit fast. **Naive users stay** — high switching costs trap them on a deteriorating platform even as trust collapses around them.

Churn by user type and intensity



Percent of each user type churned, by scenario (mean ± 95% CI).

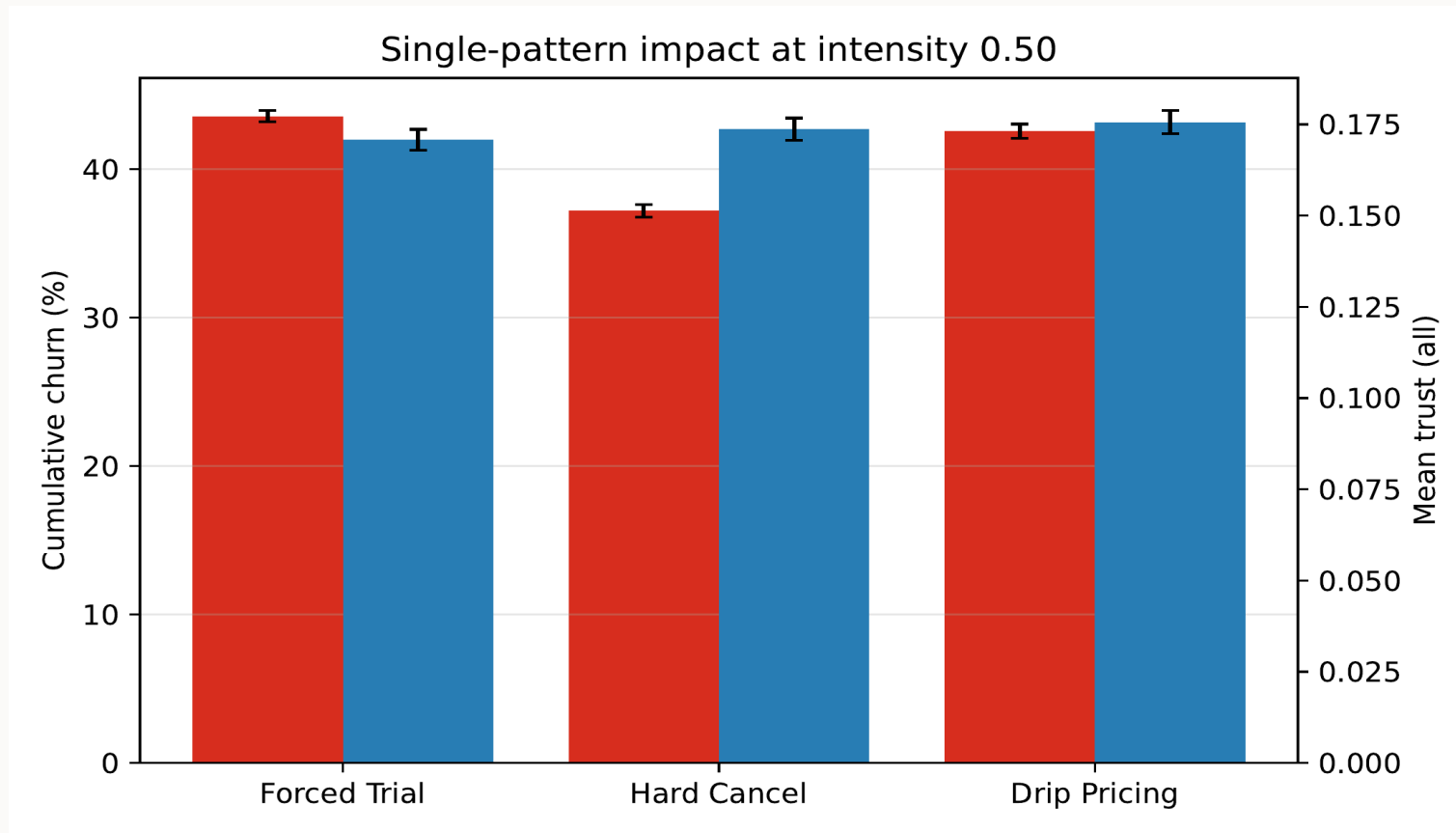
Not all dark patterns are equally harmful

TABLE 3 · SINGLE-PATTERN IMPACT AT INTENSITY 0.50 · 100 SEEDS

Metric	Forced Trial	Drip Pricing	Hard Cancel
Cumulative churn	43.5%	42.5%	37.2%
Mean trust (all)	0.171	0.176	0.174
Tipping points (of 4)	3.86	3.38	2.79

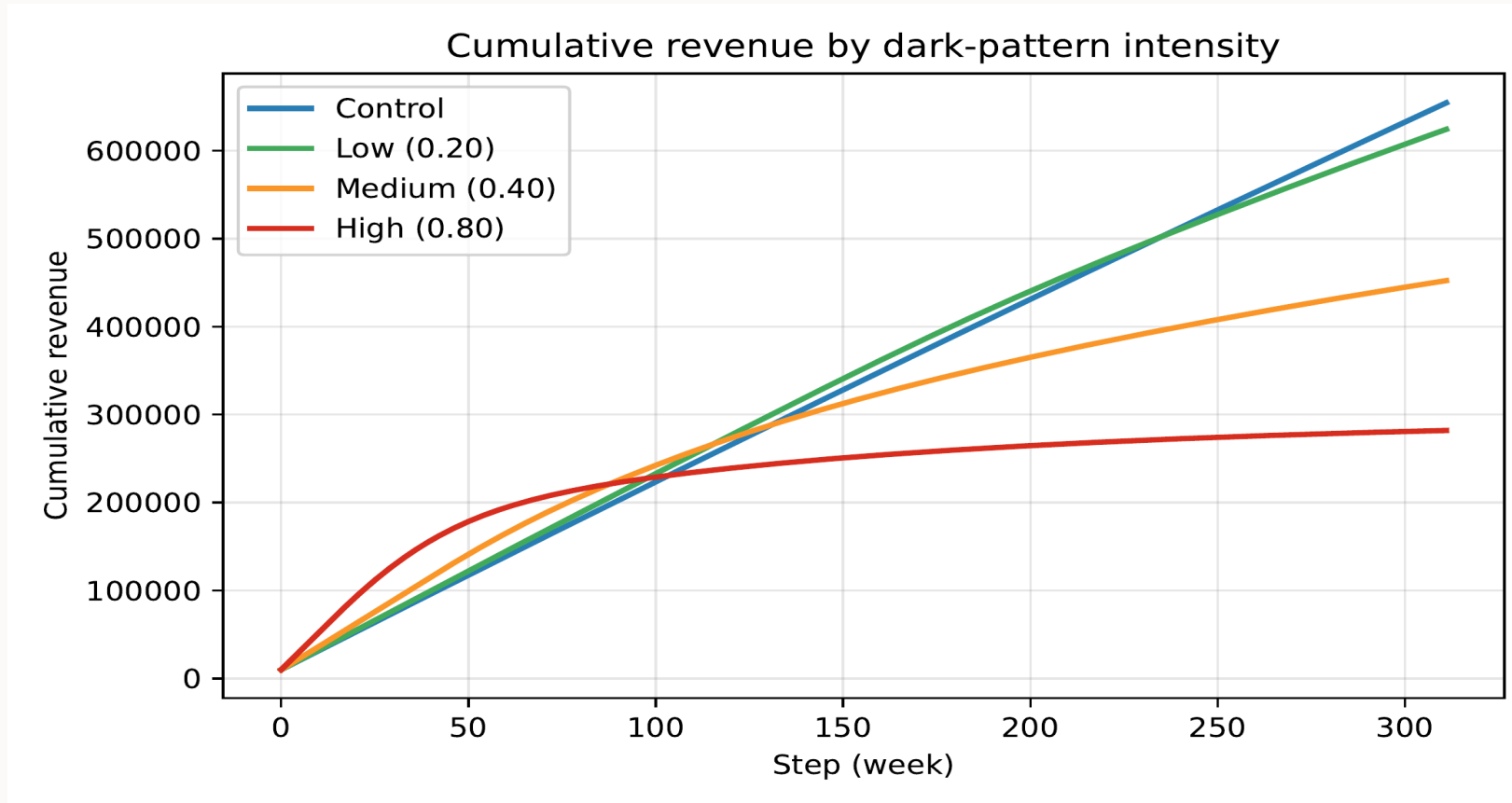
Counter-intuitive finding: Hard Cancellation is the most obstructive pattern but the *least* damaging overall — because it's the easiest to detect. Users notice fast and leave before harm accumulates. **Higher detectability shortens the exposure window and limits systemic harm.**

Per-pattern churn and residual trust



Single-pattern impact at intensity 0.50: cumulative churn and residual trust (mean $\pm$ 95% CI).

Revenue: early extraction, long-run losses



Cumulative revenue over time by dark-pattern intensity (mean $\pm$ 95% CI). Dark-pattern extraction front-loads revenue; falling reputation and churn erode it over the long run.

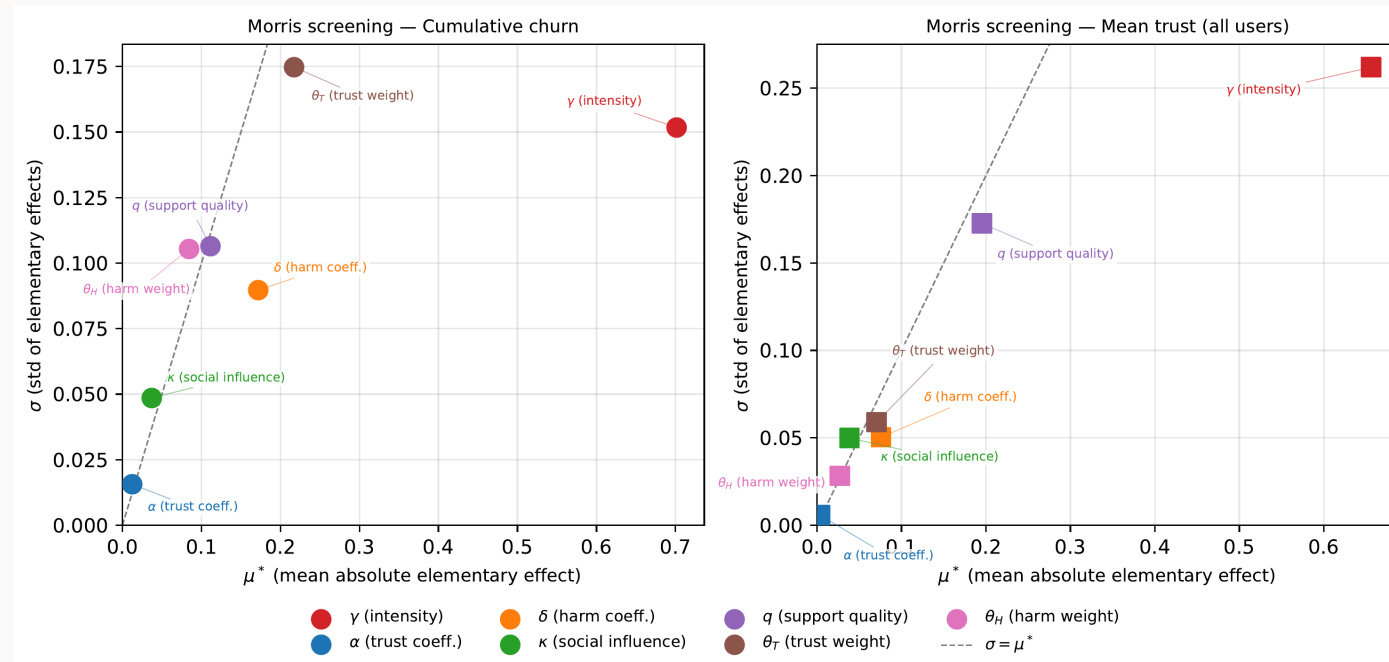
Sensitivity analysis: what actually drives the outcome?

TABLE 4 · MORRIS ELEMENTARY-EFFECTS SCREENING · 360 MODEL RUNS

Parameter	μ^* churn	μ^* trust
γ — dark pattern intensity	0.701	0.656
θ_T — churn trust weight	0.217	0.071
δ — exposure-to-harm coeff.	0.172	0.076
q — support quality	0.111	0.195
θ_H — churn harm weight	0.084	0.027
κ — social influence	0.037	0.038
α — exposure-to-trust coeff.	0.012	0.004

Intensity's effect is **3.2–3.4× larger** than any other parameter — and every signed effect points the expected direction across all 7 parameters tested simultaneously.

Robust across simultaneous parameter perturbation

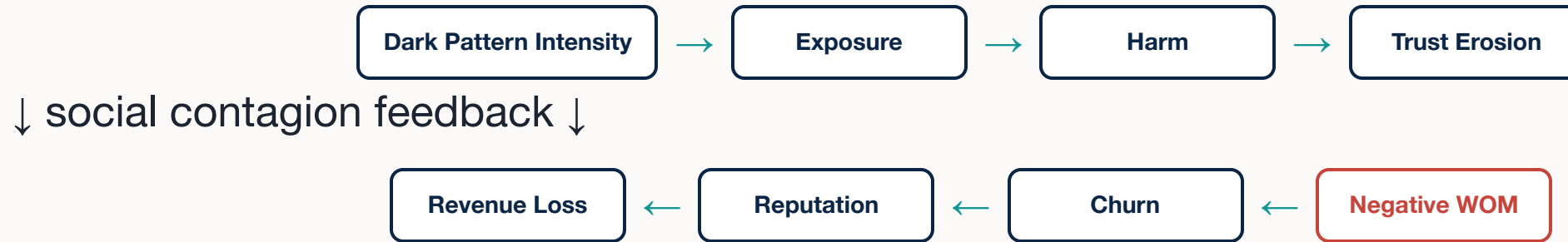


μ^* vs. σ for cumulative churn (circles) and mean trust (squares). Points far right = most influential; above the diagonal = interaction-dominated.

PART 4

Discussion

The self-reinforcing destruction loop



Negative WOM feeds back and erodes trust even among **unexposed** users (dashed loop) — accelerating every downstream stage. A strategy that looks profitable early destroys more value than it captures over the long run.

Heterogeneous vulnerability and lock-in

Who is harmed most is not who leaves.

- Naive users accumulate the **most cumulative harm** — precisely because high switching costs trap them
- Activists churn fastest but **amplify harm via WOM** before exiting
- As savvy users leave, aggregate complaint volume can **fall** — even as remaining users suffer more

A regulator watching aggregate complaint rates or reputation scores could see an **apparent improvement** at the exact moment the most vulnerable users are experiencing the worst conditions.

Vulnerability here is **situational**, not demographic — invisible to standard protected-group frameworks.

Policy implications

1. Regulate intensity, not just presence.

Non-linear, threshold-dependent harm favors graduated limits (like emission caps) over binary bans — intensity monitoring over incident-by-incident adjudication.

2. Treat dark patterns as a public harm.

Negative WOM damages users never directly exposed — an externality that individual-consent/redress frameworks aren't built to address.

3. Mandate Extractive Divergence disclosure.

A platform-economics metric platforms already track internally — a candidate "leverage ratio" for systemic-risk monitoring, ahead of reputational collapse.

Limitations

- Parameters are **literature-informed, not empirically fitted** — results are directional/qualitative, not point forecasts
- **Single platform, no competition** — a deliberate scoping choice; likely a **conservative lower bound** on real-world churn
- Does **not model** regulatory intervention, media coverage, or legal consequences
- **Static network structure** across the simulation horizon

Claims are conditional: they describe what follows *given* the modeled mechanisms — the robust part is the direction, ordering, and relative magnitude of effects, confirmed under Morris screening and alternate network topologies.

Conclusion

No safe deployment level exists within the modeled regime. Even low-intensity dark patterns cross qualitative regime boundaries and inflict persistent trust damage that is nearly invisible on a revenue dashboard.

- Harm is not contained to individual moments of manipulation — it **propagates through social networks** and **destabilizes platform economics**
- Trust Collapse triggers first and consistently → candidate **early-warning indicator**
- Extractive Divergence grounds regulatory monitoring in **economics platforms already track**

Future work: empirical calibration · multi-platform competition · adaptive regulatory dynamics (media, penalties)

Thank you

Questions & discussion

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Code, data & replication package: github.com/deting/dark-patterns-ABM